

Multiple Choice

1. **Answer:** A

Options: A) Tundra; B) Rainforest; C) Grassland; D) Chaparral; E) Savanna

Note: The correct answer is Tundra because it is characterized by cold temperatures, particularly in winter, and is often associated with vegetation such as low shrubs, mosses, and lichens, rather than extensive pine forests, which are more typical of boreal forests or taiga.

2. **Answer:** A

Options: A) an individual's phenotype; B) an individual's genotype; C) a population's phenotype; D) a species' genome; E) a species' genotype

Note: An individual's phenotype is the smallest unit that natural selection can change because natural selection acts on the observable traits of organisms, which are determined by their phenotypes, rather than directly on their genotypes or entire populations.

3. **Answer:** A

Options: A) Lactose crystallization at low temperatures causes milk to spoil; B) Milk spoils due to absorption of odors in the fridge; C) Milk spoils due to the presence of mesophiles; D) Milk spoils due to chemical reactions in the fridge; E) Thermophiles cause milk to spoil in the refrigerator

Note: Lactose crystallization at low temperatures alters the physical and chemical properties of milk, leading to spoilage as it affects texture and promotes microbial growth.

4. **Answer:** A

Options: A) RNase H; B) Helicase; C) integrase; D) DNA ligase; E) protease

Note: RNase H is crucial for the reverse transcription process in retroviruses, allowing for the conversion of RNA into DNA, which has been fundamental in developing techniques such as cloning and gene therapy in molecular biology.

5. **Answer:** E

Options: A) auxins; B) neurotransmitters; C) enzymes; D) hormones; E) pheromones

Note: Pheromones are chemical substances produced by organisms that trigger specific physiological or behavioral responses in other individuals of the same species, facilitating communication and coordination.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The statement is false because lung tissue adequate for supporting respiratory needs first evolved in early tetrapods, which include ancestors of modern amphibians, but the first vertebrates to develop true lungs were likely the early ancestors of reptiles.

7. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the sex of each child is determined by independent genetic factors, with each child having an equal probability of being male or female, regardless of the sexes of previous children.

8. **Answer:** False

Options: A) True; B) False

Note: A photosynthetic plant in iron- and magnesium-deficient soil will exhibit stunted growth and poor health because both iron and magnesium are essential nutrients that play critical roles in chlorophyll synthesis and overall plant metabolism.

9. **Answer:** True

Options: A) True; B) False

Note: Acetylcholine primarily acts as a neurotransmitter that transmits signals between nerve cells and does not typically function as an intracellular messenger within cellular signaling pathways.

10. **Answer:** False

Options: A) True; B) False

Note: The statement is false because only about 1-2% of the human genome encodes proteins, while the majority consists of non-coding regions, including regulatory elements and non-coding RNAs.

Long Form Response

11. **Answer:** Heritability is a measure of how much a trait's variation can be attributed to genetic factors, as opposed to environmental influences. In the case of abdominal bristles in *Drosophila melanogaster*, the mean number of bristles in the parent generation is 42.8, while the F_1 generation exhibits a mean of 40.6. To calculate heritability, we can use the formula $h^2 = \frac{V_g}{V_p}$, where V_g represents genetic variance and V_p is phenotypic variance. Given that the heritability value is determined to be 0.89, this indicates that a significant portion of the variation in bristle number is due to genetic differences rather than environmental factors. This high heritability suggests that selective breeding of *Drosophila* for abdominal bristle traits could effectively enhance or reduce the trait in subsequent generations, illustrating the power of genetic inheritance in shaping phenotypic characteristics.

Note: The answer correctly identifies heritability as the proportion of phenotypic variation in a trait that can be attributed to genetic differences, and it begins to outline the calculation process using the appropriate formula, emphasizing the significance of understanding both genetic and environmental contributions to traits like abdominal bristles in *Drosophila melanogaster*.

12. **Answer:** Stem reptiles hold a crucial place in the evolutionary history of modern reptiles, as they represent the earliest ancestors that exclusively evolved into contemporary reptilian species without any other descendants. This unique lineage highlights the significant adaptations and evolutionary innovations that occurred during the transition from early amniotes to the diverse array of reptiles we see today. Understanding stem reptiles allows biologists to trace the morphological and ecological changes that led to the emergence of various reptilian groups, such as lizards, snakes, and turtles. Their study provides insights into key evolutionary processes, including speciation and environmental adaptation, thereby enhancing our comprehension of reptilian evolution as a whole.

Note: correct because it emphasizes the importance of stem reptiles as the direct ancestors of modern reptiles, showcasing their evolutionary adaptations and innovations that paved the way for the diversity of reptiles seen today.